

H5: crystallography: Frieze groups, wallpaper groups, space groups

finite subgroups of SO(3)

subgroups of SO(3)	find all possible representations of SO(3) > these provide a faithful representation of SO(3) ie: there's a 1to1 correspondence between group elements and matrices we will do this in steps: 1: eigenvalues 2: element X 3: orbits 4: k formula 5: classification																																				
1: eigenvalues	given a matrix $A \in \text{SO}(3)$ its eigenvalues are 1 and $e^{\pm i\theta}$ > we have an axis of rotation $\alpha\rangle$ invariant to rotation, corresponding to eigenvalue 1																																				
2: element X	given a finite subgroup $G \subset \text{SO}(3)$ with n elements > take an element $X \neq e$ corresponding to axis of rotation $\alpha\rangle$ > define $G^\alpha \subseteq G$ the subgroup having this axis of rotation, which is finite and Abelian > corresponding eigenvalues must be of form $\theta = \pm 2\pi m/n_\alpha$ for some integer n_α > this subgroup is isomorphic to Z_{n_α} and we call the axis n_α-fold > we can assume that X has eigenvalues 1 and $e^{\pm 2\pi/n_\alpha}$																																				
3: orbits	take this n_α-fold axis of rotation and any element $Y \in G$ > then $Y \alpha\rangle$ is also an n_α fold axis of rotation nl: we see that it has eigenvalue 1 for: $(YXY^{-1})Y \alpha\rangle = Y \alpha\rangle$ consider the complete orbit associated to $\alpha\rangle$ by applying all group elements Y to it > call this orbit O_1 with associated integer $n_1 = n_\alpha$ > number of distinct elements in this orbit is $\#G/n_1$ nl: if $Y_1 \alpha\rangle = Y_2 \alpha\rangle$, $\exists m : Y_2^{-1}Y_1 = X^m$, and hence Y_1 and Y_2 give rise to the same point in the orbit iff they are related to each other by a power of X, and there are precisely n_1 distinct powers. we can repeat this for any n_j-fold axis of all elements of group G lying on the unit sphere > we end up with k orbits O_j with each $\#G/n_j$ elements > the union of all orbits is the set S of all axis of rotation on the unit sphere																																				
4: k formula	the following holds: $\sum_{i=1}^k \left(1 - \frac{1}{n_i}\right) = 2(1 - 1/n)$ proof: consider set $P = \{(\alpha\rangle, A), \alpha\rangle \in S, A \in G, A \neq I, A \alpha\rangle = \alpha\rangle\}$ > there are two ways of counting the number of elements in the set: 1: for every element in the group $A \in G$, there are two associated axis of rotation: $\pm \alpha\rangle$ > we have to exclude I, hence: $\#P = 2(\#G - 1)$ 2: to every element of an orbit O_i, we can associate $n_i - 1$ group elements in P > hence: $\#P = \sum_{i=1}^k \#O_i(n_i - 1) = \#G \sum_{i=1}^k (1 - 1/n_i)$																																				
5: classification	we have $n, n_i > 3$ and thus the only possible cases are $k=2$ and $k=3$ > otherwise the left hand side would be larger than the right hand side > we have: <table><tr><th>k</th><th>n_1</th><th>n_2</th><th>n_3</th><th>$\#G$</th><th>Name</th></tr><tr><td>2</td><td>n</td><td>n</td><td>—</td><td>n</td><td>C_n</td></tr><tr><td>3</td><td>2</td><td>2</td><td>n</td><td>$2n$</td><td>D_n</td></tr><tr><td>3</td><td>2</td><td>3</td><td>3</td><td>12</td><td>T</td></tr><tr><td>3</td><td>2</td><td>3</td><td>4</td><td>24</td><td>C</td></tr><tr><td>3</td><td>2</td><td>3</td><td>5</td><td>60</td><td>I</td></tr></table> >> there are only 5 different possible rotation groups on the unit sphere nl: the 1,2,3,4 and 6-fold rotations > rotations on a line, dihedral, tetrahedron, cube and icosahedron	k	n_1	n_2	n_3	$\#G$	Name	2	n	n	—	n	C_n	3	2	2	n	$2n$	D_n	3	2	3	3	12	T	3	2	3	4	24	C	3	2	3	5	60	I
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5.2 crystallography	
classification of wallpaper groups	We can combine rotations and reflections in 2D to form symmetry groups > there are 17 different groups > for a crystal invariant under $2\pi/n$ rotations, n can be: n=1,2,3,4,6 > we can classify these using linear algebra, in steps: 1: the lattice 2: integer matrices 3: point groups 4: symmorphic space groups 5: Bravais lattices 6: glide plane symmetries and screw axis symmetries
1: the lattice	= periodic structure of vertices, with vertices located at the coordinates: $n_1\vec{e}_1 + n_2\vec{e}_2 + n_3\vec{e}_3 \text{ with } \vec{e}_i \text{ the lattice vectors and } n_i \in \mathbb{Z}.$ denote the matrix of lattice vectors $E = (\vec{e}_1, \vec{e}_2, \vec{e}_3)$
2: integer matrices	we are searching for lattice structures with rotational and/or reflection symmetries > these are represented by 3x3 matrices O_g in the group $O(3)$ > we are looking for all subgroups of $O(3)$ for which there exists an integer representation N_g > for which there is a basis E such that: $O_g \cdot E = E \cdot N_g$ and N_g only contains integer coefficients > this is a necessary and sufficient condition for this rotation/reflection to map lattice vectors to integer linear combinations of those lattice vectors > two solutions N_g which are equivalent to each other by conjugation with an element in $GL(3, \mathbb{Z})$ correspond to the same symmetry - nl: $GL(3, \mathbb{Z})$ is the set of matrices with integer coeff for which $\det = \pm 1$ > we are only interested in the inequivalent matrix representations
3: point groups	we already classified all subgroups of $SO(3)$ and of $O(3)$ > now, we also have the restriction that the elements O_g are equivalent to an integer matrix > the eigenvalues of an integer matrix corresponding to an element with order n are: $\pm 1, \pm e^{i2\pi/n}, \pm e^{-i2\pi/n}$ the trace of these matrices has to be an integer, since the matrix is filled with integers > we have the restriction that n=1,2,3,4,6 The remaining groups are point groups, for which we have 32 of them
4: symmorphic space groups	we have reduced the problem to finding inequivalent integer representations of all rotation/reflection groups in the list of 32 point groups > there are only a few of such representations > in 3D there are 73 inequivalent integer representations N_g > these define 73 symmorphic space groups - ie: space groups for which there is one point which remains invariant
5: Bravais lattices	given a solution N_g, we can construct basis vectors E, by solving the linear eq: $O_g \cdot E = E \cdot N_g$ however, many of the solutions correspond to the same lattice E > there are 14 distinct solutions, corresponding to the Bravais lattices - ie: there are 73 representations which are inequivalent from the point of view of transformations in $SL(3, \mathbb{Z})$, but only 14 if we allow conjugation with $SL(3, \mathbb{R})$

6: glide plane symmetries and screw axis symmetries

However, there are also symmetries that don't leave any point invariant

> in 3D these are glide plane symmetries and screw axis symmetries

> we have to introduce group elements of the form:

$$(\vec{\omega}(g), g)$$

with $g \in O(3)$ labelling the rotation/reflection

$\omega(g) \in \mathbb{R}^3$ a translation vector in space

> we first do a point group operation and then a translation with vector $\omega(g)$

> for a given set of lattice vectors transforming according to one of the 73 possible integer representations N_g , the following group structure emerges:

$$(\omega(g_2), g_2) \cdot (\omega(g_1), g_1) = (\omega(g_2) + N_{g_2} \cdot \omega(g_1), g_2 \cdot g_1)$$

with the matrix N_g acting on the vector ω

> we have to rotate the sifted vector ω_1 and then add a new shift ω_2 to it

> we want that these group elements form a closed group

> we must have that the right hand side of this equation is equal to:

$$(\omega(g_2) + N_{g_2} \cdot \omega(g_1), g_2 \cdot g_1) \equiv (\omega(g_2 \cdot g_1), g_2 \cdot g_1)$$

up to integer shifts with the lattice vectors

> consistency dictates the vector equation:

$$\omega(g_2) + N_{g_2} \cdot \omega(g_1) - \omega(g_2 \cdot g_1) = 0 \quad \text{mod } Z^3$$

this is a linear set of $3n^2$ equations with n the order of the group for $3n$ unknowns

> we can solve this eq and find all inequivalent solutions ω_g